L04 ELECTRIC LIGHT GLARE CONTROL | O (MAX : 2 PT)

Intent : Minimize glare caused by electric light.

Summary : This WELL feature requires projects to manage glare by using strategies, such as calculation of glare and choosing the appropriate light fixtures for the space.

Issue : Glare is defined as excessive brightness of the light-source, excessive brightness-contrasts and excessive quantity of light[1,2] and is an integral part of lighting design. Reducing glare improves the visual experience of the occupants in the space. Glare has been associated with a host of health issues that range from visual discomfort and eye fatigue to headaches and migraines.[1,3] Studies have also shown that glare can lead to visual impairment and discomfort, which can cause accidents in the workplace. Individuals under the age of 50 are more sensitive to glare.[4] Since a substantial section of the workforce falls into this age group, it is important to address glare to avoid visual fatigue and glare-induced headaches.

Solutions : Electric lighting, the light source, type of luminaires and lighting layout can help to reduce glare.

PART 1 MANAGE GLARE FROM ELECTRIC LIGHTING (MAX : 2 PT)

For All Spaces except Industrial:

Option 1: Luminaire considerations

All luminaires within regularly occupied spaces (excluding wall wash fixtures, concealed fixtures, emergency lighting and decorative fixtures installed as specified by the manufacturer) meet one of the following requirements when measured at light output representative of regular use conditions:

a. 100% of light is emitted above the horizontal plane.

b. Classified with Unified Glare Rating (UGR) of 19 or lower.

c. Luminance that does not exceed 6,000 cd/m^2 at any angle between 45 and 90 degrees from nadir.

OR

Option 2: Space considerations

All regularly occupied spaces meet the following requirement:

a. Classified with Unified Glare Rating (UGR) of 19 or lower.

For Industrial:

Option 1: Luminaire considerations

All luminaires within regularly occupied spaces (excluding wall wash fixtures, concealed fixtures, emergency lighting and decorative fixtures installed as specified by the manufacturer) meet one of the following requirements when measured at light output representative of regular use conditions:

a. 100% of light is emitted above the horizontal plane.

b. Classified with Unified Glare Rating (UGR) of 19 or lower.

c. Luminance that does not exceed 6,000 cd/m^2 at any angle between 45 and 90 degrees from nadir.

OR

Option 2: Space considerations

All regularly occupied spaces meet the following requirement:

a. Unified Glare Rating (UGR) of 19 or lower.

REFERENCES

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